

- A Thévenin equivalent circuit consists of a voltage source in series with a resistance. A Norton equivalent circuit consists of a current source in parallel with a resistance. The Thévenin and Norton equivalent circuits are related by a source transformation.
- The parameters of the Thévenin and Norton equivalent circuits can be determined using any two of the following: (1) the open-circuit voltage at the interface, (2) the short-circuit current at the interface, and (3) the equivalent resistance of the source circuit with all sources turned off.
- The parameters of the Thévenin and Norton equivalent circuits can also be determined using circuit reduction methods or by directly solving for the source i - v relationship using node-voltage or mesh-current analysis.
- For a fixed source and an adjustable load, the maximum interface signal levels are $v_{\text{MAX}} = v_{\text{OC}} (R_L = \infty)$, $i_{\text{MAX}} = i_{\text{SC}} (R_L = 0)$, and $p_{\text{MAX}} = v_{\text{OC}} i_{\text{SC}} (R_L = R_T)$. When $R_L = R_T$, the source and load are said to be matched.
- Interface signal transfer conditions are specified in terms of the voltage, current, or power delivered to the load. The design constraints depend on the signal conditions specified and the circuit parameters that are adjustable. Some design requirements may require a two-port interface circuit. An interface design problem may have one, many, or no solutions.

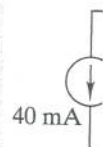


FIGURE P3-4

- 3-4 (a) Formulate node-voltage equations for the circuit in Figure P3-4.
(b) Use these equations to find v_x and i_x .



FIGURE P3-5

- 3-5 (a) Formulate node-voltage equations for the circuit in Figure P3-5.
(b) Solve for v_x and i_x when $v_s = 20 \text{ V}$.



FIGURE P3-6

- 3-6 (a) Formulate node-voltage equations for the circuit in Figure P3-6.
(b) Solve for v_x and i_x when $v_s = 10 \text{ kV}$, and $R_L = 10 \text{ k}\Omega$.

PROBLEMS

OBJECTIVE 3-1 GENERAL CIRCUIT ANALYSIS (SECT. 3-1 TO 3-2)

Given a circuit consisting of linear resistors and independent sources:

- (a) (Formulation) Write node-voltage or mesh-current equations for the circuit.
(b) (Solution) Solve the equations from part (a) for selected signal variables or input-output relationships.

Node-voltage method:

See Examples 3-1, 3-2, 3-3, 3-4, 3-5, 3-6 and Exercises 3-2, 3-3, 3-4, 3-5, 3-6

Mesh-current method:

See Examples 3-7, 3-8, 3-9 and Exercises 3-8, 3-9, 3-10

- 3-1 (a) Formulate node-voltage equations for the circuit in Figure P3-1.
(b) Use these equations to find v_x and i_x .

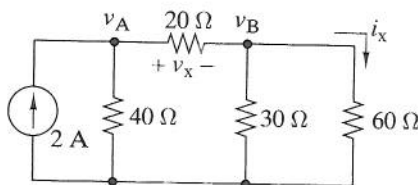


FIGURE P3-1

- 3-2 (a) Formulate node-voltage equations for the circuit in Figure P3-2.
(b) Use these equations to find v_x and i_x .

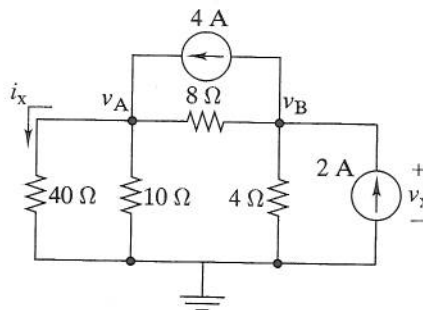


FIGURE P3-2

- 3-3 (a) Formulate node-voltage equations for the circuit in Figure P3-3.
(b) Use these equations to find v_x and i_x .

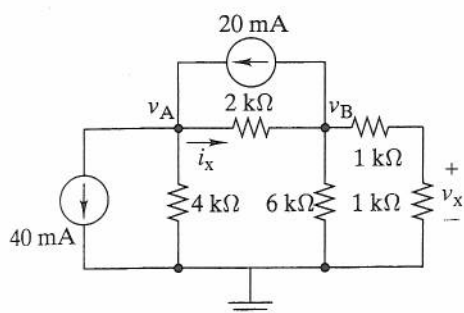


FIGURE P3-3

- 3-4 (a) Formulate node-voltage equations for the circuit in Figure P3-4.
 (b) Use these equations to find v_x and i_x .

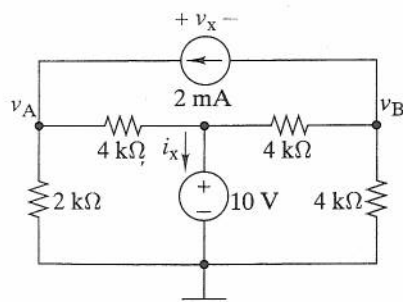


FIGURE P3-4

- 3-5 (a) Formulate node-voltage equations for the circuit in Figure P3-5.
 (b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = 10 \text{ k}\Omega$, $v_s = 20 \text{ V}$, and $i_s = 2 \text{ mA}$.

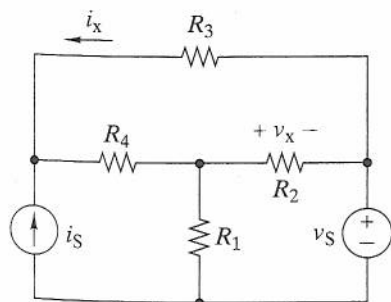


FIGURE P3-5

- 3-6 (a) Formulate node-voltage equations for the circuit in Figure P3-6.
 (b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = R_5 = 10 \text{ k}\Omega$, and $v_s = 20 \text{ V}$.

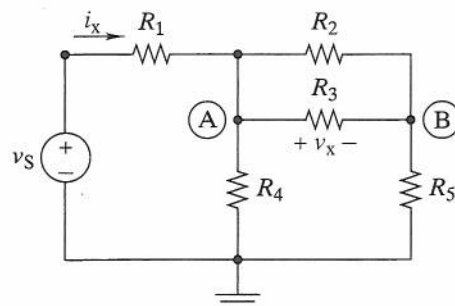


FIGURE P3-6

- 3-7 (a) Formulate node-voltage equations for the circuit in Figure P3-7.
 (b) Solve for v_A , v_B , and v_C when $R_1 = 1 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$, $R_3 = 4 \text{ k}\Omega$, $R_4 = 2 \text{ k}\Omega$, and $i_{s1} = i_{s2} = 2 \text{ mA}$.

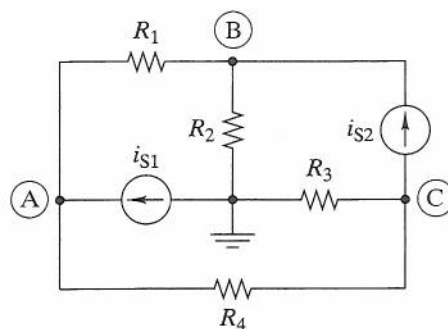


FIGURE P3-7

- 3-8 (a) Formulate node-voltage equations for the circuit in Figure P3-8.
 (b) Solve for v_x and i_x when $R_1 = R_4 = 1 \text{ k}\Omega$, $R_2 = R_3 = 250 \Omega$, $R_x = 500 \Omega$, and $v_s = 15 \text{ V}$.

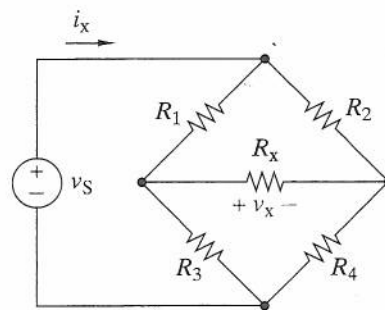


FIGURE P3-8

- 3-9 (a) Formulate mesh-current equations for the circuit in Figure P3-9.
 (b) Use these equations to find v_x and i_x .

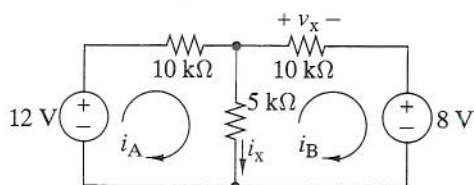


FIGURE P3-9

3-10 (a) Formulate mesh-current equations for the circuit in Figure P3-10.

(b) Use these equations to find v_x and i_x .

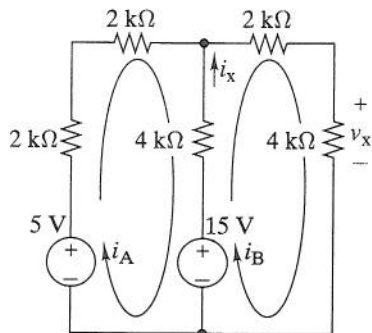


FIGURE P3-10

3-11 (a) Formulate mesh-current equations for the circuit in Figure P3-11.

(b) Use these equations to find v_x and i_x .

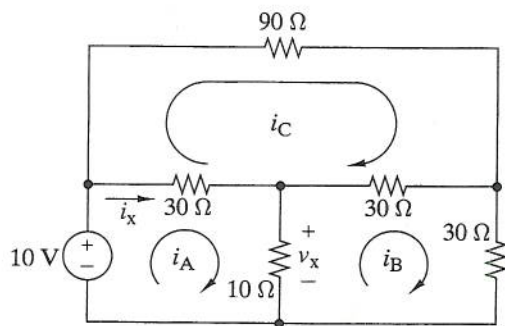


FIGURE P3-11

3-12 (a) Formulate mesh-current equations for the circuit in Figure P3-12.

(b) Solve for v_x and i_x when $R_1 = 200 \Omega$, $R_2 = 300 \Omega$, $R_3 = 50 \Omega$, $R_4 = 250 \Omega$, $R_5 = 200 \Omega$, $i_s = 50 \text{ mA}$, and $v_s = 15 \text{ V}$.

(c) Find the total power dissipated in the circuit.

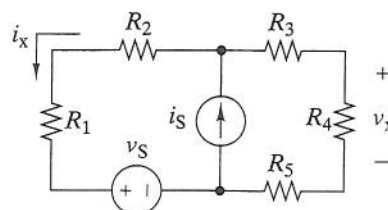


FIGURE P3-12

3-13 (a) Formulate mesh-current equations for the circuit in Figure P3-13.

(b) Solve for v_x and i_x when $R_1 = R_2 = 10 \text{ k}\Omega$, $R_3 = 2 \text{ k}\Omega$, $R_4 = 1 \text{ k}\Omega$, $i_s = 2.5 \text{ mA}$, $v_{s1} = 12 \text{ V}$, and $v_{s2} = 0.5 \text{ V}$.

(c) Find the power supplied by v_{s1} .

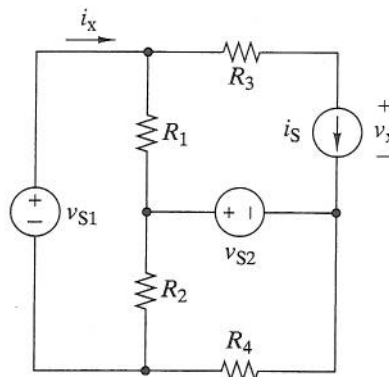


FIGURE P3-13

3-14 The circuit in Figure P3-14 seems to require two supermeshes since both current sources appear in two meshes. However, sometimes rearranging the circuit diagram will eliminate the need for a supermesh.

(a) Show that supermeshes can be avoided in Figure P3-14 by rearranging the connection of resistor R_6 .

(b) Formulate mesh-current equations for the modified circuit as redrawn in part (a).

(c) Solve for v_x when $R_1 = R_2 = R_3 = R_4 = 2 \text{ k}\Omega$, $R_5 = R_6 = 1 \text{ k}\Omega$, $i_{s1} = 40 \text{ mA}$, and $i_{s2} = 20 \text{ mA}$.

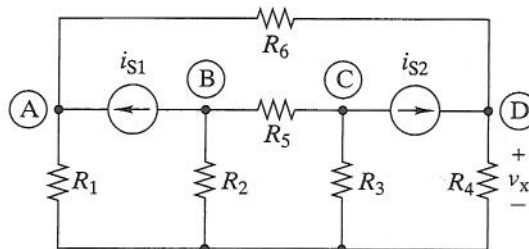
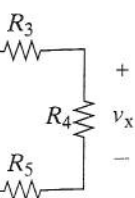


FIGURE P3-14

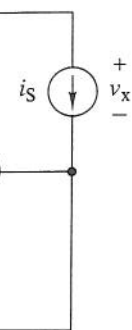
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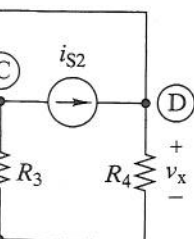


equations for the circuit
 $R_2 = 10 \text{ k}\Omega$, $R_3 = 2 \text{ k}\Omega$,
 $V = 12 \text{ V}$, and $v_{s2} = 0.5 \text{ V}$.



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 section of resistor R_6 .
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$R_3 = R_4 = 2 \text{ k}\Omega$, $R_5 =$
 $s_2 = 20 \text{ mA}$.



- 3-15 (a) Formulate mesh-current equations for the circuit in Figure P3-15.
 (b) Use these equations to find v_x and i_x .
 (c) Find the total power delivered to the resistors.

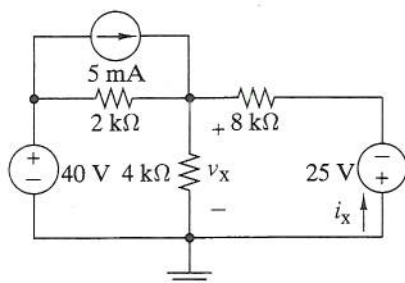


FIGURE P3-15

- 3-16 (a) Formulate mesh-current equations for the circuit in Figure P3-16.
 (b) Use these equations to find the input resistance.

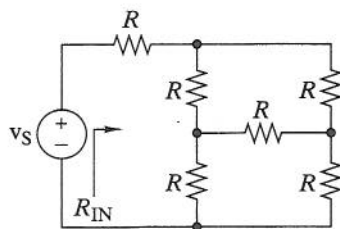


FIGURE P3-16

- 3-17 In Figure P3-17 all of the resistors are $1 \text{ k}\Omega$. The voltage at node C is found to be $v_C = -2 \text{ V}$ when node B is connected to ground. Find the node voltages v_A and v_D , and the mesh currents i_A and i_B .

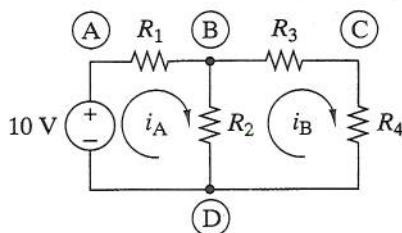


FIGURE P3-17

- 3-18 In Figure P3-17 all of the resistors are $1 \text{ k}\Omega$. The voltage at node A is observed to be $v_A = 8 \text{ V}$ when node C is connected to ground. Find the node voltages v_B and v_D , and the mesh currents i_A and i_B .

- 3-19 Find the node voltages v_A and v_B in Figure P3-19.

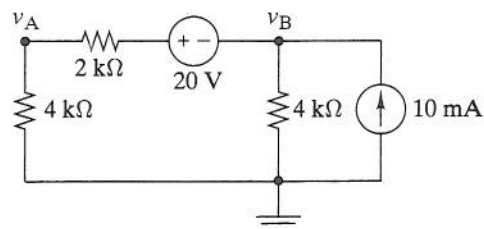


FIGURE P3-19

- 3-20 Find the mesh currents i_A , i_B , and i_C in Figure P3-20

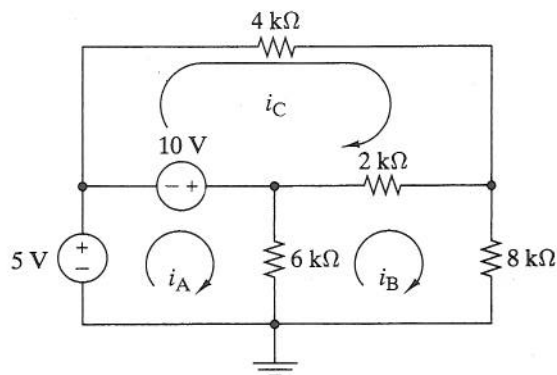


FIGURE P3-20

OBJECTIVE 3-2 LINEARITY PROPERTIES (SECT. 3-3)

- (a) Given a circuit containing linear resistors and one independent source, use the proportionality principle to find selected signal variables.
 (b) Given a circuit containing linear resistors and two or more independent sources, use the superposition principle to find selected signal variables.
 See Examples 3-10, 3-11, 3-12 and Exercises 3-11, 3-13, 3-14, 3-15

- 3-21 Find the proportionality constant $K = v_O/i_S$ for the circuit in Figure P3-21.

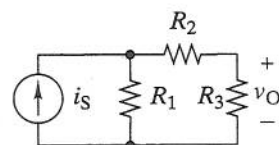


FIGURE P3-21

3-64 Design the interface circuit in Figure P3-63 so that 25 mW is delivered to the 1-k Ω load resistor.

3-65 Use the resistor array in Figure P3-65 to design a voltage divider with $K = v_2/v_1 \geq 0.6$ and an input resistance close to 100 Ω .

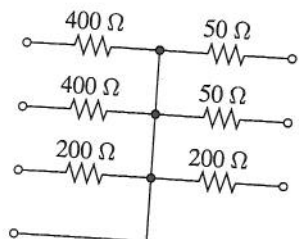


FIGURE P3-65

3-66 Select R_L in Figure P3-66 so that the power delivered to the load is 400 mW.

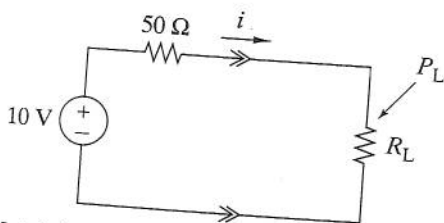


FIGURE P3-66

3-67 Design the interface circuit in Figure P3-67 so that $R_{IN} = 100 \Omega$ and the current delivered to the 50- Ω load is $i = 50$ mA.

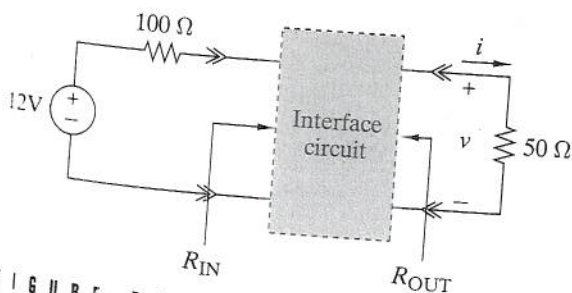


FIGURE P3-67

3-68 Design the interface circuit in Figure P3-67 so that $R_{OUT} = 50 \Omega$ and the voltage delivered to the 50- Ω load is $v = 2$ V.

3-69 The circuit in Figure P3-69 has a source resistance of 75 Ω and a load resistance of 600 Ω . Design the interface circuit so that the input resistance is $R_{IN} = 75 \Omega \pm 10\%$ and the output resistance is $R_{OUT} = 600 \Omega \pm 10\%$.

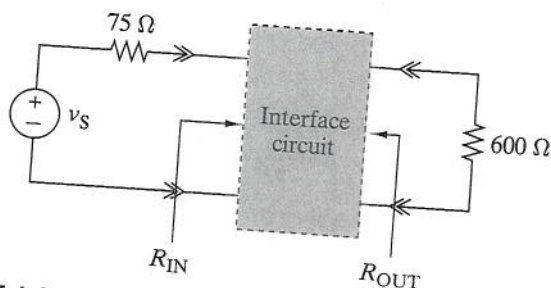


FIGURE P3-69

3-70 It is claimed that both interface circuits in Figure 3-70 will deliver $v = 4$ V to the 75- Ω load. Verify this claim. Which interface circuit consumes the least power? Which has an output resistance that best matches the 75- Ω load?

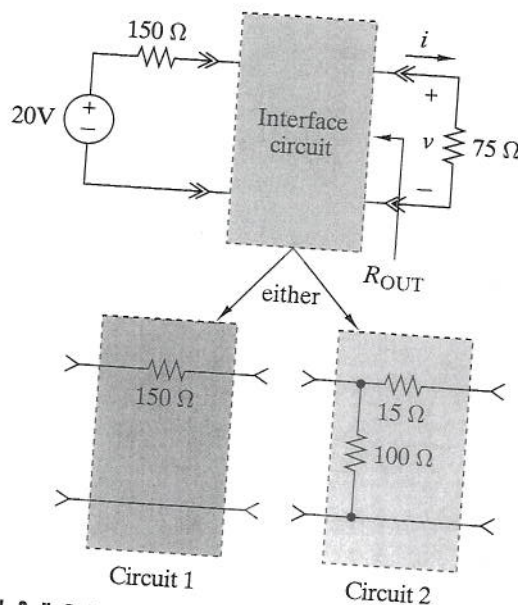


FIGURE P3-70

INTEGRATING PROBLEMS

3-71 Composite Circuit Analysis
The $i-v$ characteristic of network N in Figure P3-71 is $i = (5 \times 10^{-3}v - 0.5)$ A. Find the voltage at node A.

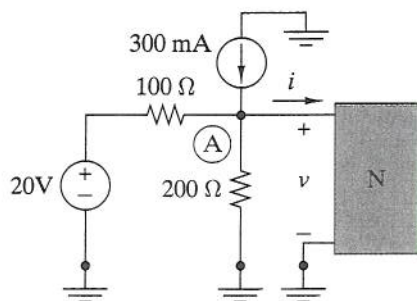


FIGURE P3-71

3-72 **Attenuation Analysis**

In Figure P3-72 a two-port attenuator connects a $600\text{-}\Omega$ source to a $600\text{-}\Omega$ load. Find the power delivered to the load in terms of v_s . Remove the attenuator and find the power delivered to the load when the source is directly connected to the load. By what fraction does the attenuator reduce the power delivered to the $600\text{-}\Omega$ load? Express the fraction in dB.

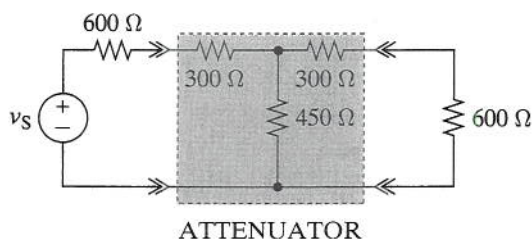


FIGURE P3-72

3-73 **Interface Circuit Design**

Using no more than three $50\text{-}\Omega$ resistors, design the interface circuit in Figure P3-73 so that $v \leq 4\text{ V}$ and $i \leq 100\text{ mA}$ regardless of the value of R_L .

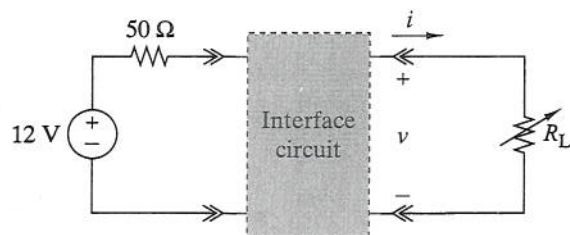


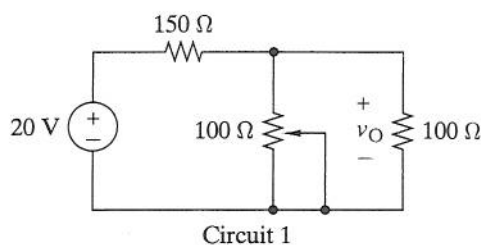
FIGURE P3-73

3-74 **Battery Design**

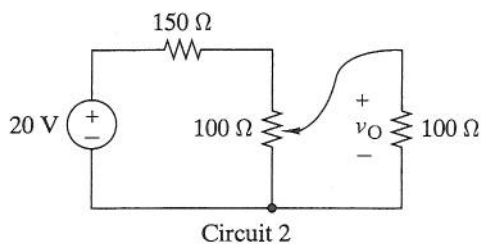
A satellite requires a battery with an open-circuit voltage $v_{OC} = 36\text{ V}$ and a Thévenin resistance $R_T \leq 10\text{ }\Omega$. The battery is to be constructed using series and parallel combinations of one of two types of cells. The first type has $v_{OC} = 9\text{ V}$, $R_T = 4\text{ }\Omega$, and a weight of 40 grams. The second type has $v_{OC} = 4\text{ V}$, $R_T = 0.5\text{ }\Omega$, and a weight of 15 grams. Design a minimum-weight battery that meets the open-circuit voltage and Thévenin resistance requirements.

3-75 **Design Evaluation**

A requirement exists for a circuit to deliver 0 to 5 V to a $100\text{-}\Omega$ load from a 20-V source rated at 2.5 W . Two proposed circuits are shown in Figure P3-75. Which one would you choose and why?



Circuit 1



Circuit 2

FIGURE P3-75